

Podman Commands Cheat Sheet

A concise reference guide for managing containers and images in **Podman**, designed for data science and reproducible research workflows (e.g., BIOS 512 or 611 projects).

Image Management

Command	Description
<code>podman build -t <name> .</code>	Build an image from a <code>Containerfile</code> in the current directory. Use <code>-t</code> to tag/name it.
<code>podman images</code>	List all locally available images.
<code>podman rmi <image-id></code>	Remove a specific image.
<code>podman tag <old-name> <new-name></code>	Rename or retag an image.
<code>podman inspect <image-id></code>	View detailed metadata for an image (environment, layers, etc.).

Container Lifecycle

Command	Description
<code>podman run -d -p 8787:8787 -e PASSWORD=rstudio -v \$(pwd):/home/rstudio/ project:Z <image></code>	Run a container in detached mode, exposing RStudio on port 8787 with mounted project files.
<code>podman ps</code>	List running containers.
<code>podman ps -a</code>	List all containers (running or stopped).
<code>podman stop <container-id></code>	Stop a running container.
<code>podman start <container-id></code>	Start a stopped container.
<code>podman restart <container-id></code>	Restart a container.
<code>podman rm <container-id></code>	Remove a stopped container.
<code>podman exec -it <container-id> bash</code>	Open an interactive bash shell inside a running container.

Volume & File Management

Command	Description
<code>podman volume ls</code>	List named volumes.
<code>podman volume inspect <volume-name></code>	Show volume details (paths, usage).
<code>podman volume rm <volume-name></code>	Delete an unused volume.
<code>podman cp <container-id>:/path/in/container ./path/on/host</code>	Copy files from container to host.
<code>podman cp ./path/on/host <container-id>:/path/in/container</code>	Copy files from host to container.

Troubleshooting & Inspection

Command	Description
<code>podman logs <container-id></code>	View logs from a running container.
<code>podman inspect <container-id></code>	Inspect configuration and metadata.
<code>podman stats</code>	Show live CPU/memory stats for all running containers.
<code>podman top <container-id></code>	Show processes running inside a container.

Cleanup Commands

Command	Description
<code>podman stop \$(podman ps -aq)</code>	Stop all containers.
<code>podman rm \$(podman ps -aq)</code>	Remove all stopped containers.
<code>podman rmi \$(podman images -aq)</code>	Delete all images.
<code>podman system prune -a</code>	Remove all unused containers, images, and volumes. <i>(Be careful!)</i>
<code>podman system df</code>	Show disk space usage by Podman resources.

Development & Debugging

Command	Description
<code>podman run -it --rm <image> bash</code>	Run container interactively and auto-remove when finished.
<code>podman port <container-id></code>	Show port mappings.
<code>podman inspect --format '{{.NetworkSettings.IPAddress}}' <container-id></code>	Display container's IP address.

Version & Info

Command	Description
<code>podman version</code>	Display Podman version.
<code>podman info</code>	Show system-level information (storage, network).
<code>podman help</code>	List all available commands.
<code>podman <command> --help</code>	Get details for a specific command.

Example: BIOS512 RStudio Container

```
# Build the image
podman build -t bios512-wine .

# Run RStudio container
podman run -d -p 8787:8787 -e PASSWORD=rstudio -v $(pwd):/home/rstudio/
project:Z bios512-wine

# View running containers
podman ps

# Stop and remove the container
podman stop <container-id>
podman rm <container-id>
```